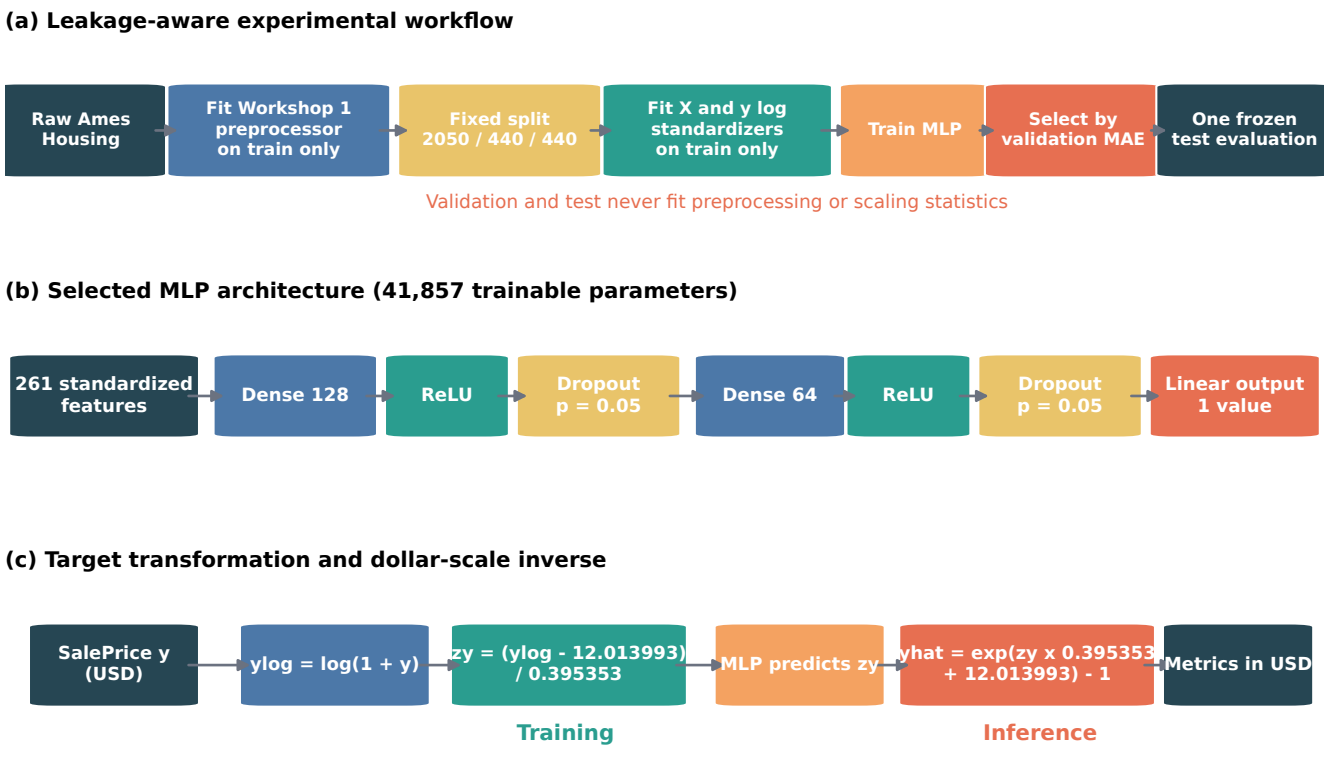
(a) Leakage-aware experimental workflow
Raw Ames Housing
Fit Workshop 1 preprocessor on train only
Fixed split 2050 / 440 / 440
Fit X and y log standardizers on train only
Train MLP
Select by validation MAE
One frozen test evaluation
Validation and test never fit preprocessing or scaling statistics
(b) Selected MLP architecture (41,857 trainable parameters)
261 standardized features
Dense 128
ReLU
Dropout p = 0.05
Dense 64
ReLU
Dropout p = 0.05
Linear output 1 value
(c) Target transformation and dollar-scale inverse
SalePrice y (USD)
ylog = log(1 + y)
zy = (ylog - 12.013993) / 0.395353
MLP predicts zy
yhat = exp(zy x 0.395353 + 12.013993) - 1
Metrics in USD
Training
Inference